

Suggested FYP topics for 2026-27

by

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Project code :	WLOY1
Project title :	Persistent 3D Spatial Memory for Video-Only Spatial Reasoning Problem statement.
No. of students :	2
Description of project :	Persistent 3D spatial reasoning from video remains a challenging problem because most vision systems process frames independently and lack a mechanism to retain long-term spatial information. As a result, they struggle to reason about object locations, spatial relationships, and scene structure when objects become occluded or move out of view. This project focuses on developing a persistent 3D spatial memory framework that reconstructs and maintains a consistent representation of the environment from video-only observations. Students will explore methods that integrate video-based 3D reconstruction with memory mechanisms to accumulate spatial knowledge over time. The project involves designing models that build and update 3D spatial representations from video streams and evaluating their ability to support spatial reasoning tasks such as object localization, scene understanding, and long-horizon reasoning.
Industrial collaboration :	No.

Project code :	WLOY 2
Project title :	Multimodal Understanding and Generation for Scientific Data Enhancement
No. of students :	2
Description of project :	Multimodal understanding and generation have shown great potential in modeling complex cross-modal relationships across text, images, and structured scientific data. By learning semantic alignment, feature representation, and controllable generation from multi-source scientific data, it can effectively expand limited scientific datasets, improve data quality, and alleviate data scarcity issues. It is contemplated that multimodal understanding and generation techniques can be systematically applied to scientific data enhancement to boost the performance of scientific machine learning models.
Industrial collaboration :	No.

Project code :	WLOY 3
Project title :	Unsupervised Change Detection in Science and Engineering
No. of students :	2
Description of project :	Change detection is a critical analytical tool across various scientific and engineering disciplines. In fields like remote sensing and astronomy, vast amounts of observational data are continuously collected over time, making manual annotation highly impractical. This project focuses on developing robust

	unsupervised change detection algorithms to identify significant spatial-temporal changes without relying on expensive, labeled datasets. Students will explore unsupervised deep learning to automatically detect critical variation. The project entails model design, implementation, and performance evaluation on real-world images.
Industrial collaboration :	No.

Project code :	WLOY 4
Project title :	Large Language Models for understanding cell Biology
No. of students :	1
Description of project :	Although large language models have shown strong capabilities in learning biological sequences and textual knowledge, they often lack explicit supervision from structured biological networks, limiting their ability to capture the causal logic underlying gene regulation. To address this limitation, we fine-tune a large language model at the parameter level using gene interaction information from authoritative biological databases—such as regulatory relationships, physical interactions, and co-expression patterns—as knowledge supervision signals, enabling the model to truly internalize the structural semantics and causal logic of gene networks.
Industrial collaboration :	No.